



**Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam –
603 110**

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of CSE

Assignment

Regulations – R2021

Degree & Branch	BE CSE			Semester	V
Subject Code & Name	UCS 2501 Computer Networks				
Academic Year	2023-2024 ODD	Batch	2021-2025	Due Date	11.10.2023

Solve the following problems using the knowledge you gained from CO1 & CO2 of the course.

1	Network with bandwidth of 20Mbps can pass only an average of 14,000 frames per minute with each frame carrying an average of 20,000 bits. What is the throughput of this network?	(CO1, K3)	5
2	What is the propagation time if the distance between the two points is 18,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.	(CO1, K3)	5
3	What are the propagation time and the transmission time for a 5.5-KB (kilobyte) message (an email) if the bandwidth of the network is 2 Gbps? Assume that the distance between the sender and the receiver is 19,000 km and that light travels at 2.4×10^8 m/s.	(CO1, K3)	5
4	If the bandwidth of the channel is 15Kbps, how long does it take to send a frame of 200,000 bits out of this device?	(CO1, K3)	5
5	Station 1 and 2 transmit pin 1, Station 3 transmits 0 and Station 4 is silent. Find the encoded data on the common channel using CDMA and decode and find how Station 1 listens to Station 2 and Station 3 listens to Station 4. Assign the codes for Station 1,2,3,4 in order. (CO2)	(CO2, K3)	10
6	The data 11011001 is being sent across the network. Use hamming code technique to encode the value. Introduce error in the 6th bit and find the error on the receiving end. (CO2)	(CO2, K3)	10

Course Incharge

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